

# flspcell\_typ\_1p20V\_25C Library

---

| Cell Groups |
|-------------|
| DFRBPQ      |

# DFRBPQ



*flspcell\_typ\_1p20V\_25C Cell Library: Process  
flspcell\_typ\_1p20V\_25C, Voltage 1.20, Temp 25.00*

---

## Truth Table

| INPUT |   |         | OUTPUT |
|-------|---|---------|--------|
| CLK   | D | RESET_B | Q      |
| R     | 0 | 1       | 0      |
| R     | 1 | 1       | 1      |
| x     | x | 0       | 0      |
| x     | x | 1       | IQ     |

## Footprint

| Cell Name           | Area     |
|---------------------|----------|
| sg13g2_dfrbpq1arx_1 | 48.98880 |

## Pin Capacitance Information

| Cell Name           | Pin Cap(pf) |         |         | Max Cap(pf) |
|---------------------|-------------|---------|---------|-------------|
|                     | CLK         | D       | RESET_B | Q           |
| sg13g2_dfrbpq1arx_1 | 0.02178     | 0.01503 | 0.02686 | 0.30000     |

## Leakage Information

| Cell Name           | Leakage(pW) |            |            |
|---------------------|-------------|------------|------------|
|                     | Min.        | Avg        | Max.       |
| sg13g2_dfrbpq1arx_1 | 2547.28000  | 2841.54000 | 3186.70000 |

## Delay Information

Delay(ns) to Q rising :

| Cell Name           | Timing Arc(Dir) | Delay(ns) |          |                |          |          |                |          |          |                |
|---------------------|-----------------|-----------|----------|----------------|----------|----------|----------------|----------|----------|----------------|
|                     |                 | Slew(ns)  | Load(pf) | First          | Slew(ns) | Load(pf) | Mid            | Slew(ns) | Load(pf) | Last           |
| sg13g2_dfrbpq1arx_1 | CLK->Q (RR)     | 0.01860   | 0.00100  | <b>0.38016</b> | 0.32940  | 0.06480  | <b>0.65900</b> | 2.50740  | 0.30000  | <b>1.55037</b> |

Delay(ns) to Q falling :

| Cell Name           | Timing Arc(Dir) | Delay(ns) |          |                |          |          |                |          |          |                |
|---------------------|-----------------|-----------|----------|----------------|----------|----------|----------------|----------|----------|----------------|
|                     |                 | Slew(ns)  | Load(pf) | First          | Slew(ns) | Load(pf) | Mid            | Slew(ns) | Load(pf) | Last           |
| sg13g2_dfrbpq1arx_1 | CLK->Q (RF)     | 0.01860   | 0.00100  | <b>0.41111</b> | 0.32940  | 0.06480  | <b>0.68193</b> | 2.50740  | 0.30000  | <b>1.41063</b> |
|                     | RESET_B->Q (FF) | 0.01860   | 0.00100  | <b>0.48423</b> | 0.32940  | 0.06480  | <b>0.79232</b> | 2.50740  | 0.30000  | <b>1.69847</b> |

## Constraint Information

Constraints(ns) for CLK rising :

| Cell Name           | Timing Check    | Ref Pin(trans) | Constraint(ns) |              |                |                |              |                |                |              |                |
|---------------------|-----------------|----------------|----------------|--------------|----------------|----------------|--------------|----------------|----------------|--------------|----------------|
|                     |                 |                | Input Slew(ns) | Ref Slew(ns) | First          | Input Slew(ns) | Ref Slew(ns) | Mid            | Input Slew(ns) | Ref Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | min_pulse_width | CLK ()         | 0.01860        | 0.00000      | <b>0.11826</b> | 1.26300        | 0.00000      | <b>2.08496</b> | 2.50740        | 0.00000      | <b>4.13818</b> |

Constraints(ns) for CLK falling :

| Cell Name           | Timing Check    | Ref Pin(trans) | Constraint(ns) |              |                |                |              |                |                |              |                |
|---------------------|-----------------|----------------|----------------|--------------|----------------|----------------|--------------|----------------|----------------|--------------|----------------|
|                     |                 |                | Input Slew(ns) | Ref Slew(ns) | First          | Input Slew(ns) | Ref Slew(ns) | Mid            | Input Slew(ns) | Ref Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | min_pulse_width | CLK ()         | 0.01860        | 0.00000      | <b>0.12466</b> | 1.26300        | 0.00000      | <b>2.08496</b> | 2.50740        | 0.00000      | <b>4.13818</b> |

Constraints(ns) for D rising :

| Cell Name           | Timing Check | Ref Pin(trans) | Constraint(ns) |              |                 |                |              |                 |                |              |                 |
|---------------------|--------------|----------------|----------------|--------------|-----------------|----------------|--------------|-----------------|----------------|--------------|-----------------|
|                     |              |                | Input Slew(ns) | Ref Slew(ns) | First           | Input Slew(ns) | Ref Slew(ns) | Mid             | Input Slew(ns) | Ref Slew(ns) | Last            |
| sg13g2_dfrbpq1arx_1 | hold         | CLK (R)        | 0.01860        | 0.01860      | <b>-0.05135</b> | 1.26300        | 1.26300      | <b>-0.19698</b> | 2.50740        | 2.50740      | <b>-0.25383</b> |
|                     | setup        | CLK (R)        | 0.01860        | 0.01860      | <b>0.45970</b>  | 1.26300        | 1.26300      | <b>0.55586</b>  | 2.50740        | 2.50740      | <b>0.59031</b>  |

Constraints(ns) for D falling :

| Cell Name           | Timing Check | Ref Pin(trans) | Constraint(ns) |              |                 |                |              |                 |                |              |                 |
|---------------------|--------------|----------------|----------------|--------------|-----------------|----------------|--------------|-----------------|----------------|--------------|-----------------|
|                     |              |                | Input Slew(ns) | Ref Slew(ns) | First           | Input Slew(ns) | Ref Slew(ns) | Mid             | Input Slew(ns) | Ref Slew(ns) | Last            |
| sg13g2_dfrbpq1arx_1 | hold         | CLK (R)        | 0.01860        | 0.01860      | <b>-0.02445</b> | 1.26300        | 1.26300      | <b>-0.15920</b> | 2.50740        | 2.50740      | <b>-0.25383</b> |
|                     | setup        | CLK (R)        | 0.01860        | 0.01860      | <b>0.50371</b>  | 1.26300        | 1.26300      | <b>0.59904</b>  | 2.50740        | 2.50740      | <b>0.65819</b>  |

Constraints(ns) for RESET\_B rising :

| Cell Name           | Timing Check | Ref Pin(trans) | Constraint(ns) |              |                 |                |              |                 |                |              |                 |
|---------------------|--------------|----------------|----------------|--------------|-----------------|----------------|--------------|-----------------|----------------|--------------|-----------------|
|                     |              |                | Input Slew(ns) | Ref Slew(ns) | First           | Input Slew(ns) | Ref Slew(ns) | Mid             | Input Slew(ns) | Ref Slew(ns) | Last            |
| sg13g2_dfrbpq1arx_1 | recovery     | CLK (R)        | 0.01860        | 0.01860      | <b>0.11981</b>  | 1.26300        | 1.26300      | <b>0.27793</b>  | 2.50740        | 2.50740      | <b>0.38370</b>  |
|                     | removal      | CLK (R)        | 0.01860        | 0.01860      | <b>-0.10759</b> | 1.26300        | 1.26300      | <b>-0.26714</b> | 2.50740        | 2.50740      | <b>-0.37189</b> |

Constraints(ns) for RESET\_B falling :

| Cell Name           | Timing Check    | Ref Pin(trans) | Constraint(ns) |              |                |                |              |                |                |              |                |
|---------------------|-----------------|----------------|----------------|--------------|----------------|----------------|--------------|----------------|----------------|--------------|----------------|
|                     |                 |                | Input Slew(ns) | Ref Slew(ns) | First          | Input Slew(ns) | Ref Slew(ns) | Mid            | Input Slew(ns) | Ref Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | min_pulse_width | RESET_B ()     | 0.01860        | 0.00000      | <b>0.10864</b> | 1.26300        | 0.00000      | <b>2.08496</b> | 2.50740        | 0.00000      | <b>4.13818</b> |

## Power Information

Internal switching power(pJ) to Q rising :

| Cell Name           | Input | Power(pJ) |          |                |          |          |                |          |          |                |
|---------------------|-------|-----------|----------|----------------|----------|----------|----------------|----------|----------|----------------|
|                     |       | Slew(ns)  | Load(pf) | First          | Slew(ns) | Load(pf) | Mid            | Slew(ns) | Load(pf) | Last           |
| sg13g2_dfrbpq1arx_1 | CLK   | 0.01860   | 0.00100  | <b>0.13998</b> | 0.32940  | 0.06480  | <b>0.13858</b> | 2.50740  | 0.30000  | <b>0.16293</b> |

Internal switching power(pJ) to Q falling :

| Cell Name           | Input   | Power(pJ) |          |                |          |          |                |          |          |                |
|---------------------|---------|-----------|----------|----------------|----------|----------|----------------|----------|----------|----------------|
|                     |         | Slew(ns)  | Load(pf) | First          | Slew(ns) | Load(pf) | Mid            | Slew(ns) | Load(pf) | Last           |
| sg13g2_dfrbpq1arx_1 | CLK     | 0.01860   | 0.00100  | <b>0.15661</b> | 0.32940  | 0.06480  | <b>0.15823</b> | 2.50740  | 0.30000  | <b>0.18528</b> |
|                     | RESET_B | 0.01860   | 0.00100  | <b>0.14528</b> | 0.32940  | 0.06480  | <b>0.14529</b> | 2.50740  | 0.30000  | <b>0.15426</b> |

Passive power(pJ) for CLK rising :

| Cell Name           | Power(pJ) |                |          |                |          |                |
|---------------------|-----------|----------------|----------|----------------|----------|----------------|
|                     | Slew(ns)  | First          | Slew(ns) | Mid            | Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | 0.01860   | <b>0.03497</b> | 0.32940  | <b>0.03462</b> | 2.50740  | <b>0.05884</b> |

Passive power(pJ) for CLK falling :

| Cell Name           | Power(pJ) |                |          |                |          |                |
|---------------------|-----------|----------------|----------|----------------|----------|----------------|
|                     | Slew(ns)  | First          | Slew(ns) | Mid            | Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | 0.01860   | <b>0.06285</b> | 0.32940  | <b>0.06266</b> | 2.50740  | <b>0.08835</b> |

Passive power(pJ) for CLK rising (conditional):

| Cell Name           | When                         | Power(pJ) |                |          |                |          |                |
|---------------------|------------------------------|-----------|----------------|----------|----------------|----------|----------------|
|                     |                              | Slew(ns)  | First          | Slew(ns) | Mid            | Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | (D *<br>RESET_B *<br>Q)      | 0.01860   | <b>0.02484</b> | 0.32940  | <b>0.02444</b> | 2.50740  | <b>0.04892</b> |
|                     | (D *<br>!RESET_B *<br>* !Q)  | 0.01860   | <b>0.03497</b> | 0.32940  | <b>0.03462</b> | 2.50740  | <b>0.05884</b> |
|                     | (!D *<br>RESET_B *<br>!Q)    | 0.01860   | <b>0.03632</b> | 0.32940  | <b>0.03608</b> | 2.50740  | <b>0.06028</b> |
|                     | (!D *<br>!RESET_B *<br>* !Q) | 0.01860   | <b>0.04182</b> | 0.32940  | <b>0.04140</b> | 2.50740  | <b>0.06563</b> |

Passive power(pJ) for CLK falling (conditional):

| Cell Name           | When                         | Power(pJ) |                |          |                |          |                |
|---------------------|------------------------------|-----------|----------------|----------|----------------|----------|----------------|
|                     |                              | Slew(ns)  | First          | Slew(ns) | Mid            | Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | (D *<br>RESET_B *<br>Q)      | 0.01860   | <b>0.06218</b> | 0.32940  | <b>0.06195</b> | 2.50740  | <b>0.08757</b> |
|                     | (D *<br>RESET_B *<br>!Q)     | 0.01860   | <b>0.06285</b> | 0.32940  | <b>0.06266</b> | 2.50740  | <b>0.08835</b> |
|                     | (D *<br>!RESET_B *<br>* !Q)  | 0.01860   | <b>0.03037</b> | 0.32940  | <b>0.03040</b> | 2.50740  | <b>0.05527</b> |
|                     | (!D *<br>RESET_B *<br>Q)     | 0.01860   | <b>0.18569</b> | 0.32940  | <b>0.17980</b> | 2.50740  | <b>0.20402</b> |
|                     | (!D *<br>RESET_B *<br>!Q)    | 0.01860   | <b>0.02670</b> | 0.32940  | <b>0.02673</b> | 2.50740  | <b>0.05159</b> |
|                     | (!D *<br>!RESET_B *<br>* !Q) | 0.01860   | <b>0.02365</b> | 0.32940  | <b>0.02368</b> | 2.50740  | <b>0.04853</b> |

Passive power(pJ) for D rising :

| Cell Name           | Power(pJ) |                |          |                |          |                |
|---------------------|-----------|----------------|----------|----------------|----------|----------------|
|                     | Slew(ns)  | First          | Slew(ns) | Mid            | Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | 0.01860   | <b>0.05511</b> | 0.32940  | <b>0.05431</b> | 2.50740  | <b>0.06021</b> |

Passive power(pJ) for D falling :

| Cell Name           | Power(pJ) |                |          |                |          |                |
|---------------------|-----------|----------------|----------|----------------|----------|----------------|
|                     | Slew(ns)  | First          | Slew(ns) | Mid            | Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | 0.01860   | <b>0.05432</b> | 0.32940  | <b>0.05369</b> | 2.50740  | <b>0.06315</b> |

Passive power(pJ) for D rising (conditional):

| Cell Name           | When              | Power(pJ) |                |          |                |          |                |
|---------------------|-------------------|-----------|----------------|----------|----------------|----------|----------------|
|                     |                   | Slew(ns)  | First          | Slew(ns) | Mid            | Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | CLK               | 0.01860   | <b>0.05281</b> | 0.32940  | <b>0.05210</b> | 2.50740  | <b>0.06090</b> |
|                     | (!CLK * RESET_B)  | 0.01860   | <b>0.08992</b> | 0.32940  | <b>0.08911</b> | 2.50740  | <b>0.09807</b> |
|                     | (!CLK * !RESET_B) | 0.01860   | <b>0.05511</b> | 0.32940  | <b>0.05431</b> | 2.50740  | <b>0.06021</b> |

Passive power(pJ) for D falling (conditional):

| Cell Name           | When              | Power(pJ) |                |          |                |          |                |
|---------------------|-------------------|-----------|----------------|----------|----------------|----------|----------------|
|                     |                   | Slew(ns)  | First          | Slew(ns) | Mid            | Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | CLK               | 0.01860   | <b>0.05432</b> | 0.32940  | <b>0.05369</b> | 2.50740  | <b>0.06315</b> |
|                     | (!CLK * RESET_B)  | 0.01860   | <b>0.07171</b> | 0.32940  | <b>0.07085</b> | 2.50740  | <b>0.08063</b> |
|                     | (!CLK * !RESET_B) | 0.01860   | <b>0.04168</b> | 0.32940  | <b>0.04099</b> | 2.50740  | <b>0.04722</b> |

Passive power(pJ) for RESET\_B rising :

| Cell Name           | Power(pJ) |                |          |                |          |                |
|---------------------|-----------|----------------|----------|----------------|----------|----------------|
|                     | Slew(ns)  | First          | Slew(ns) | Mid            | Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | 0.01860   | <b>0.01140</b> | 0.32940  | <b>0.01112</b> | 2.50740  | <b>0.01811</b> |

Passive power(pJ) for RESET\_B falling :

| Cell Name           | Power(pJ) |                |          |                |          |                |
|---------------------|-----------|----------------|----------|----------------|----------|----------------|
|                     | Slew(ns)  | First          | Slew(ns) | Mid            | Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | 0.01860   | <b>0.02629</b> | 0.32940  | <b>0.02399</b> | 2.50740  | <b>0.03612</b> |

Passive power(pJ) for RESET\_B rising (conditional):

| Cell Name           | When             | Power(pJ) |                |          |                |          |                |
|---------------------|------------------|-----------|----------------|----------|----------------|----------|----------------|
|                     |                  | Slew(ns)  | First          | Slew(ns) | Mid            | Slew(ns) | Last           |
| sg13g2_dfrbpq1arx_1 | (CLK * D * !Q)   | 0.01860   | <b>0.01140</b> | 0.32940  | <b>0.01112</b> | 2.50740  | <b>0.01811</b> |
|                     | (CLK * !D * !Q)  | 0.01860   | <b>0.00550</b> | 0.32940  | <b>0.00558</b> | 2.50740  | <b>0.00558</b> |
|                     | (!CLK * D * !Q)  | 0.01860   | <b>0.04617</b> | 0.32940  | <b>0.04513</b> | 2.50740  | <b>0.05495</b> |
|                     | (!CLK * !D * !Q) | 0.01860   | <b>0.00895</b> | 0.32940  | <b>0.00901</b> | 2.50740  | <b>0.00901</b> |

Passive power(pJ) for RESET\_B falling (conditional):

| Cell Name           | When             | Power(pJ) |                 |          |                 |          |                 |
|---------------------|------------------|-----------|-----------------|----------|-----------------|----------|-----------------|
|                     |                  | Slew(ns)  | First           | Slew(ns) | Mid             | Slew(ns) | Last            |
| sg13g2_dfrbpq1arx_1 | (CLK * D * !Q)   | 0.01860   | <b>0.17292</b>  | 0.32940  | <b>0.17025</b>  | 2.50740  | <b>0.19217</b>  |
|                     | (CLK * !D * !Q)  | 0.01860   | <b>-0.00408</b> | 0.32940  | <b>-0.00471</b> | 2.50740  | <b>-0.00493</b> |
|                     | (!CLK * D * !Q)  | 0.01860   | <b>0.02629</b>  | 0.32940  | <b>0.02399</b>  | 2.50740  | <b>0.03612</b>  |
|                     | (!CLK * !D * !Q) | 0.01860   | <b>-0.00754</b> | 0.32940  | <b>-0.00799</b> | 2.50740  | <b>-0.00813</b> |